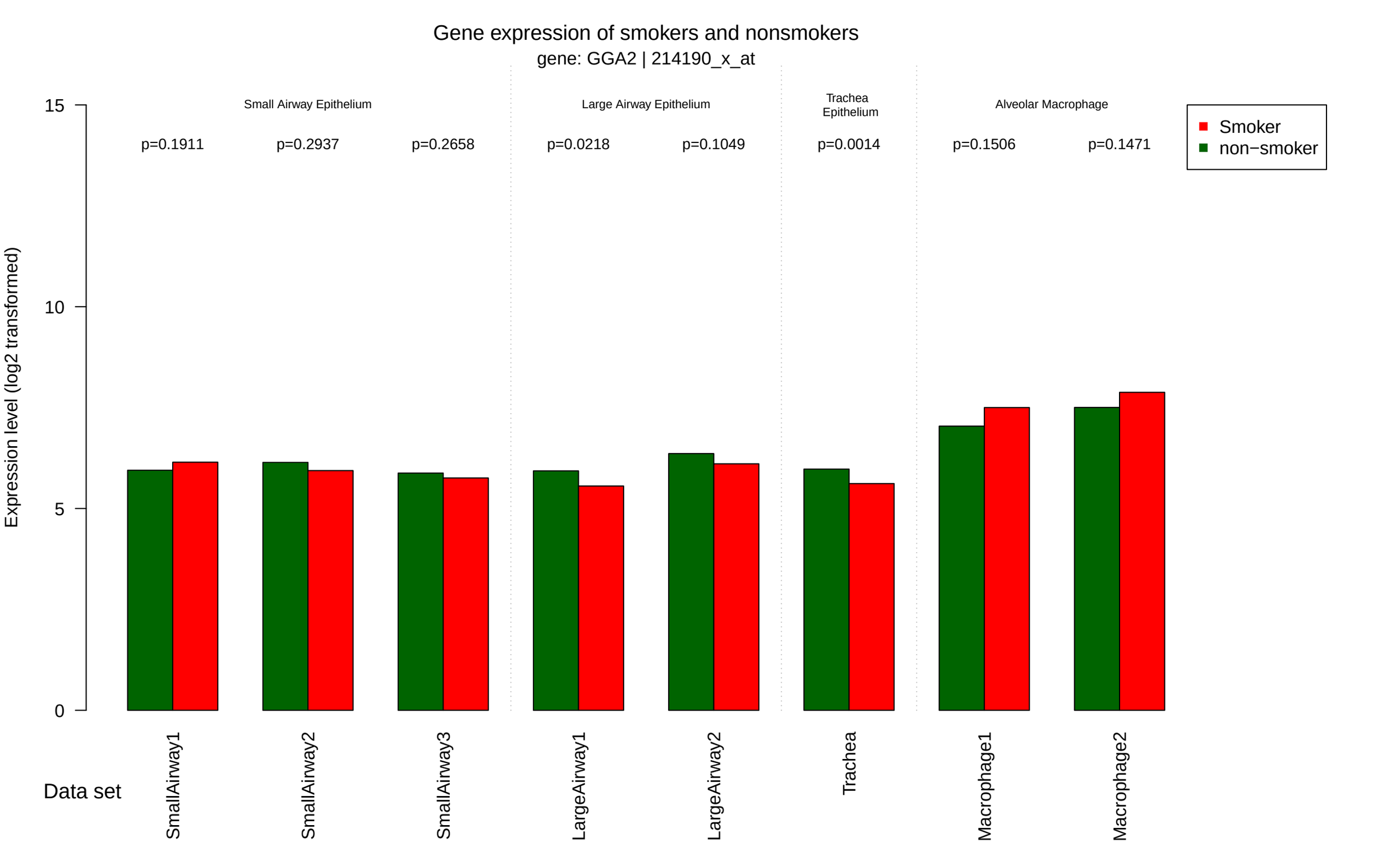
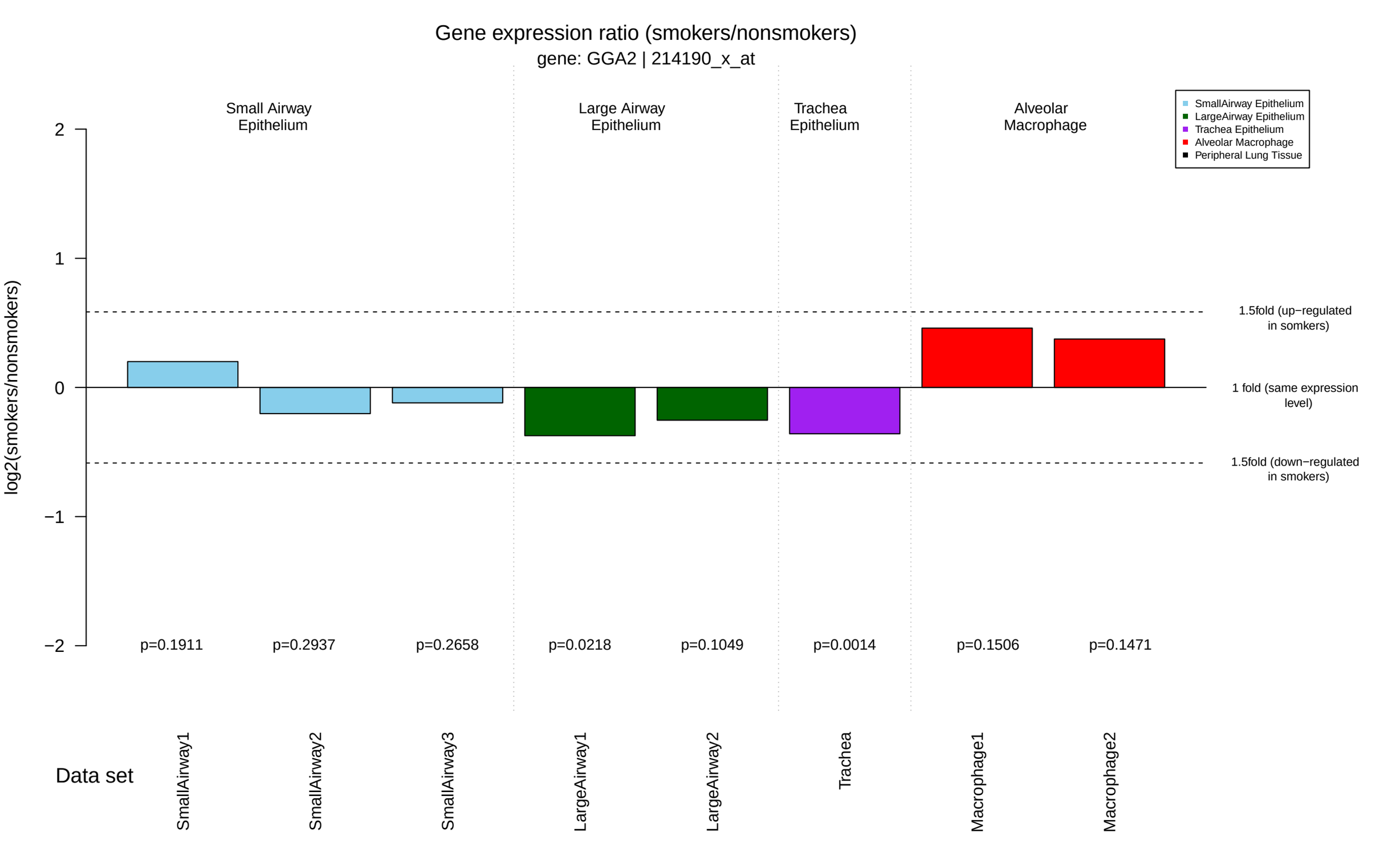
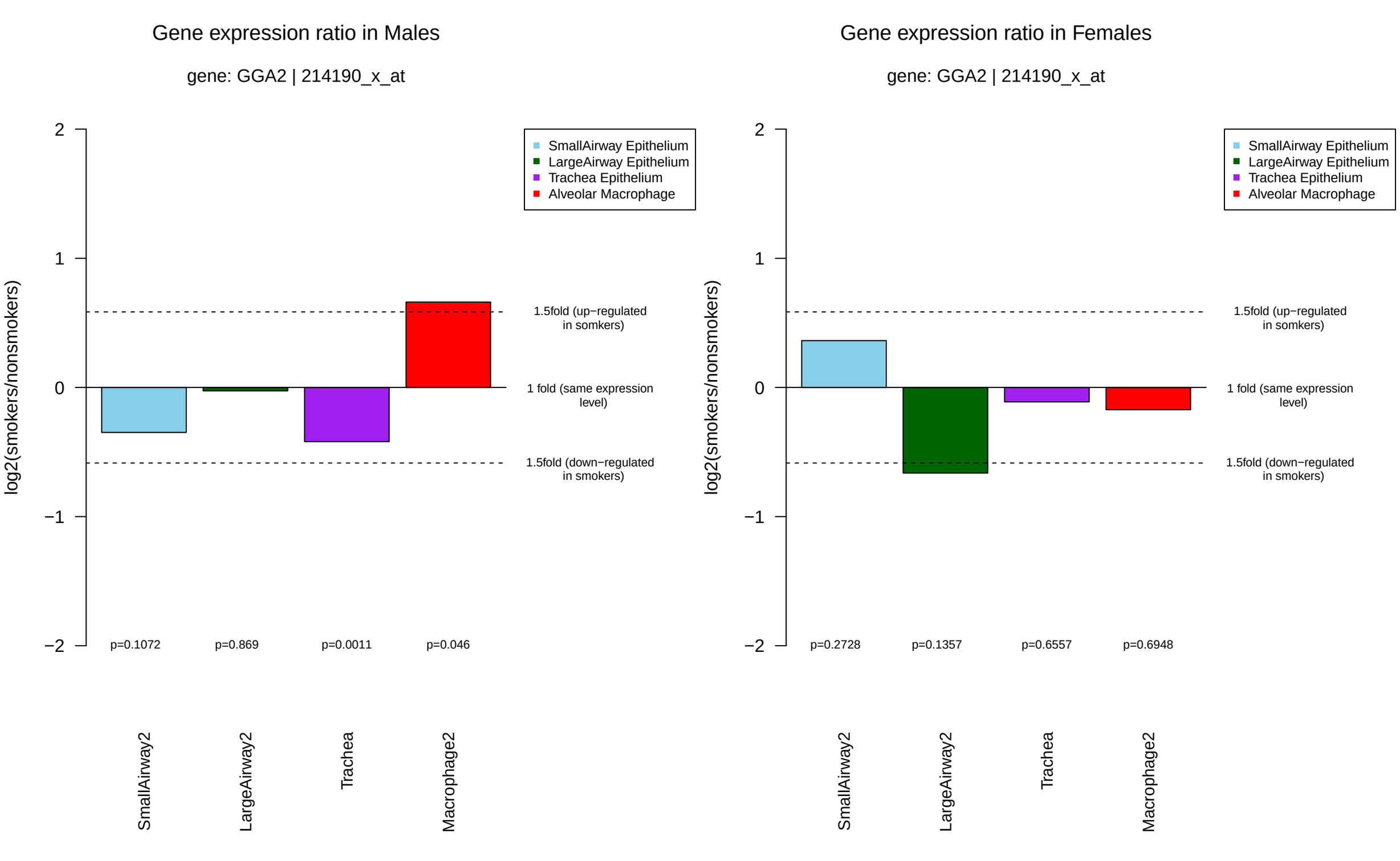
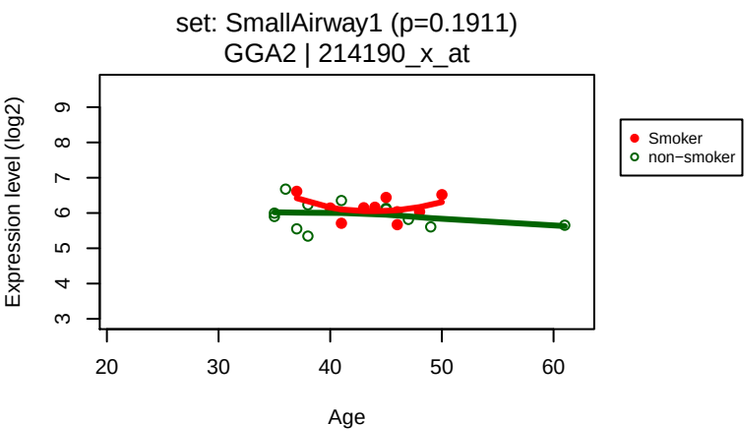
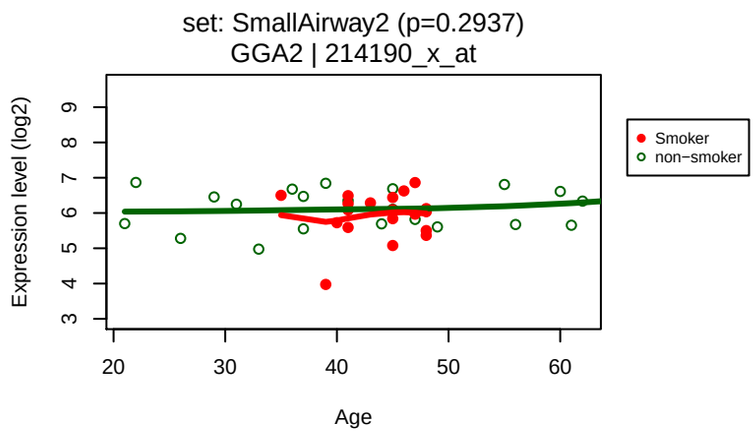
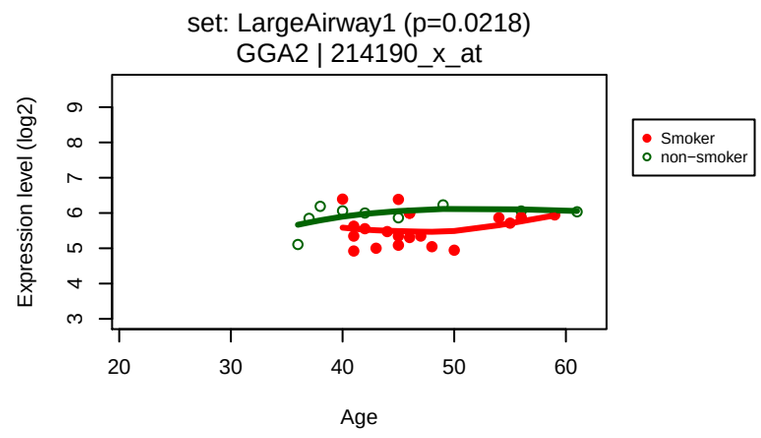

Gene expression of smokers and nonsmokers
gene: GGA2 | 214190_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1911
p=0.2937
p=0.2658
p=0.0218
p=0.1049
p=0.0014
p=0.1506
p=0.1471
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

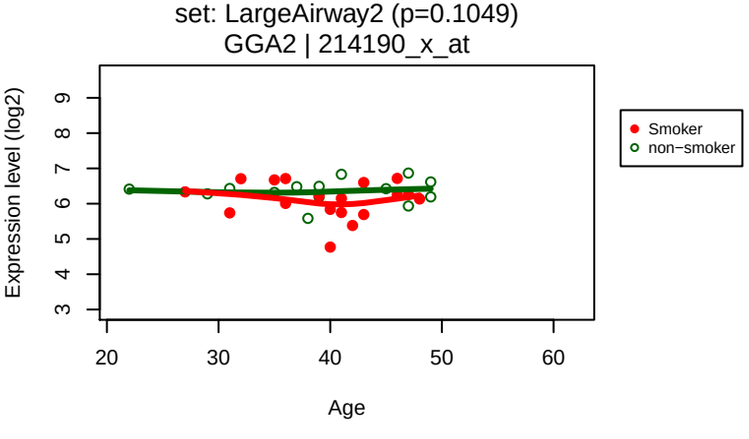

Gene expression ratio (smokers/nonsmokers)
gene: GGA2 | 214190_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1911
p=0.2937
p=0.2658
p=0.0218
p=0.1049
p=0.0014
p=0.1506
p=0.1471
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

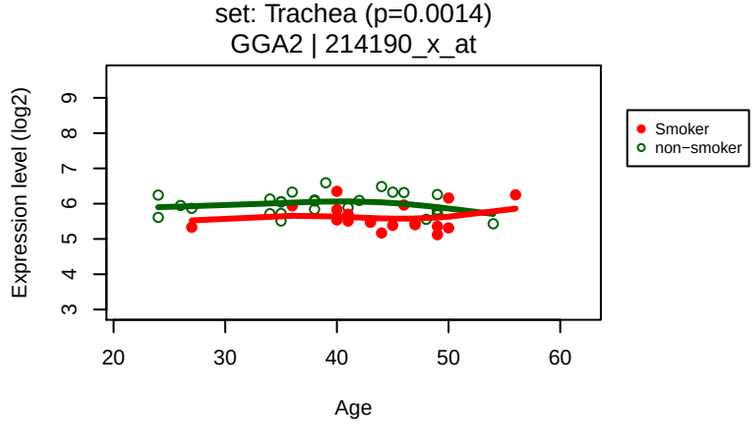

Gene expression ratio in Males
gene: GGA2 | 214190_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1072
p=0.869
p=0.0011
p=0.046
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GGA2 | 214190_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2728
p=0.1357
p=0.6557
p=0.6948
SmallAirway2
LargeAirway2
Trachea
Macrophage2

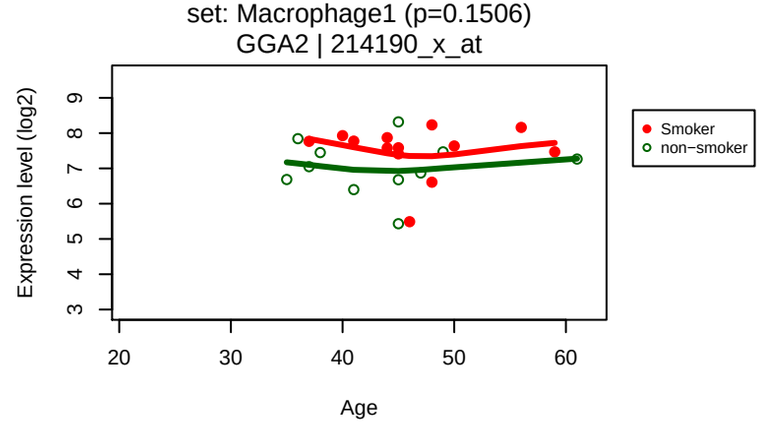

set: SmallAirway1 (p=0.1911)
GGA2 | 214190_x_at
Expression level (log2)
Age
Smoker
non−smoker

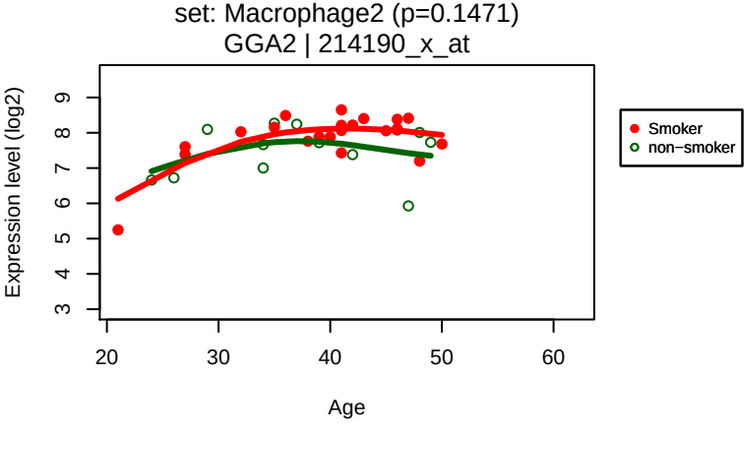

set: SmallAirway2 (p=0.2937)
GGA2 | 214190_x_at
Expression level (log2)
Age
Smoker
non−smoker

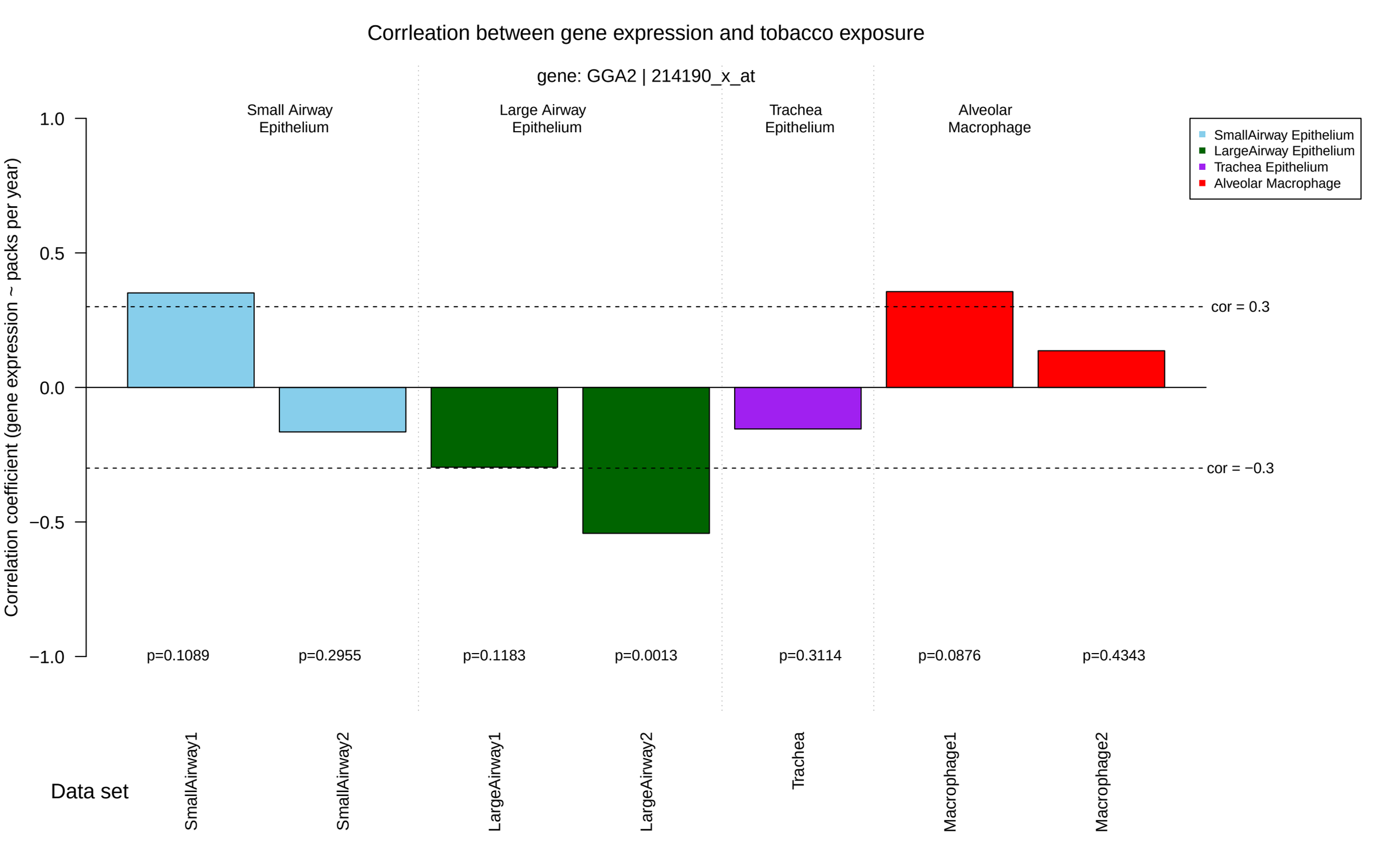

set: LargeAirway1 (p=0.0218)
GGA2 | 214190_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1049)
GGA2 | 214190_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0014)
GGA2 | 214190_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1506)
GGA2 | 214190_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1471)
GGA2 | 214190_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GGA2 | 214190_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1089
p=0.2955
p=0.1183
p=0.0013
p=0.3114
p=0.0876
p=0.4343
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2